🔥 One-Page Analog Verification Interview Cheat Sheet (2026)

Purpose: Fast recall during interviews — what to run, why, and how to justify it.
Audience: Analog / AMS / Verification Engineers (0–5 yrs)

PRE-LAYOUT (DESIGN INTENT)

- DC OP → Check biasing, saturation, headroom
- AC → Gain, PM, GBW, PSRR, CMRR
- Transient → Slew rate, settling, ringing
- Noise → Thermal + 1/f, input-referred
- Linearity → THD / IMD (ADC, DAC, RF)

PVT & CORNERS

- Process: TT, SS, FF, FS, SF
- Voltage: VDD ±10%
- Temperature: −40°C to 125°C
- Goal: Functional robustness across extremes

MONTE CARLO & YIELD

- MC (Process + Mismatch) → σ, offset, gain spread
- Run ≥1000 samples
- Worst-Case Corner + MC = Signoff condition
- Quote yield: 3σ = 99.73%, 6σ = 99.99966%

AGING & RELIABILITY

- NBTI / PBTI / HCI → Vth drift
- Aging reduces gm → gain ↓, bandwidth ↓
- EM / IR → metal & supply integrity

MIXED-SIGNAL (AMS)

- Verilog-AMS co-simulation
- Real-number modeling
- UVM-AMS assertions (lock time, conversion done)

POST-LAYOUT

- RC extraction (xRC / StarRC)
- Post-layout AC/Tran/Noise
- Expect ~10–20% BW degradation

🔥 INTERVIEW POWER STATEMENTS

1. "I always sign off at worst-case PVT + Monte Carlo."
2. "Noise is checked only after AC stability is ensured."
3. "Post-layout bandwidth drop is mainly parasitic C."
4. "Aging directly impacts gm, hence gain and phase margin."